3 **(a)** A satellite of mass m orbits a planet of mass M in a circular orbit, with orbital radius r.

Show that its kinetic energy E_K and gravitational potential energy E_P are related by

$$E_K = -\frac{E_P}{2}.$$

[2]

(b) The variation of gravitational potential ϕ with distance r from the centre of Jupiter is shown in Fig. 3.1.

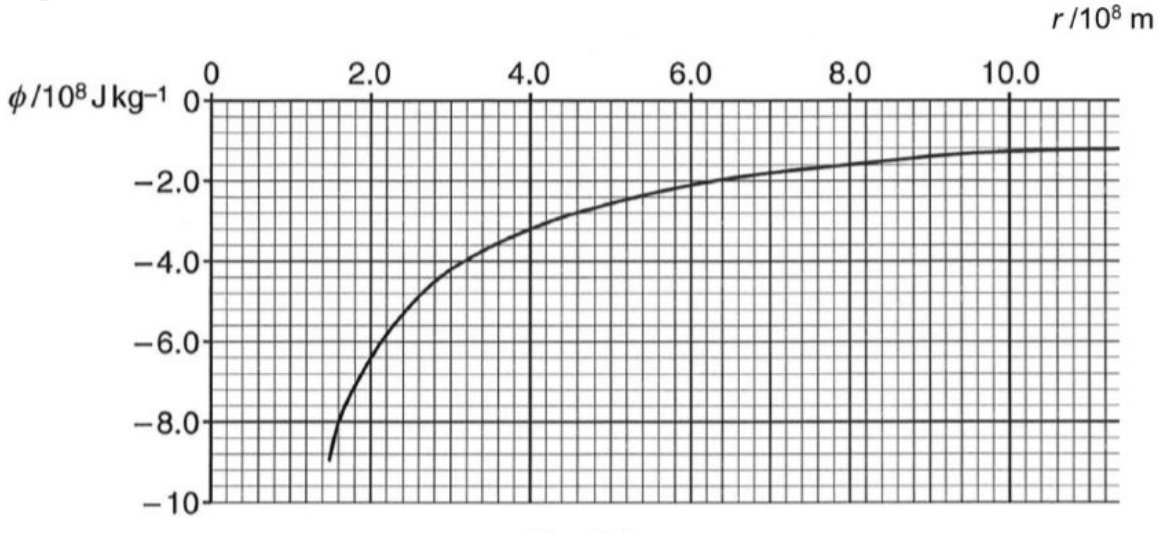


Fig. 3.1

A satellite orbiting Jupiter has a mass of 8.93×10^{22} kg and orbital radius 4.0×10^8 m.

Determine the total energy of the satellite.

total energy = ... J [3]

(c) Two identical isolated uniform spheres X and Y each have radius R. The centres of the spheres are separated by distance L, as shown in Fig. 3.2.

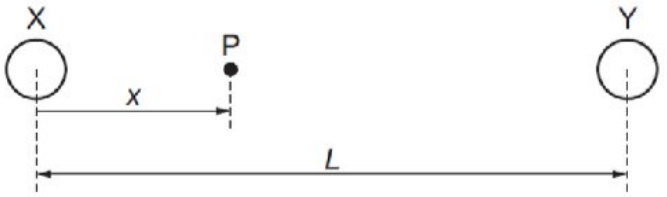


Fig. 3.2

Point P lies on the line joining the centres of X and Y, and is at a variable displacement x from the centre of sphere X.

The gravitational field strength at the surface of each sphere is g_o.

On Fig. 3.3, sketch the variation with x of the gravitational field strength g at point P between $x = R$ and $x = L - R$.

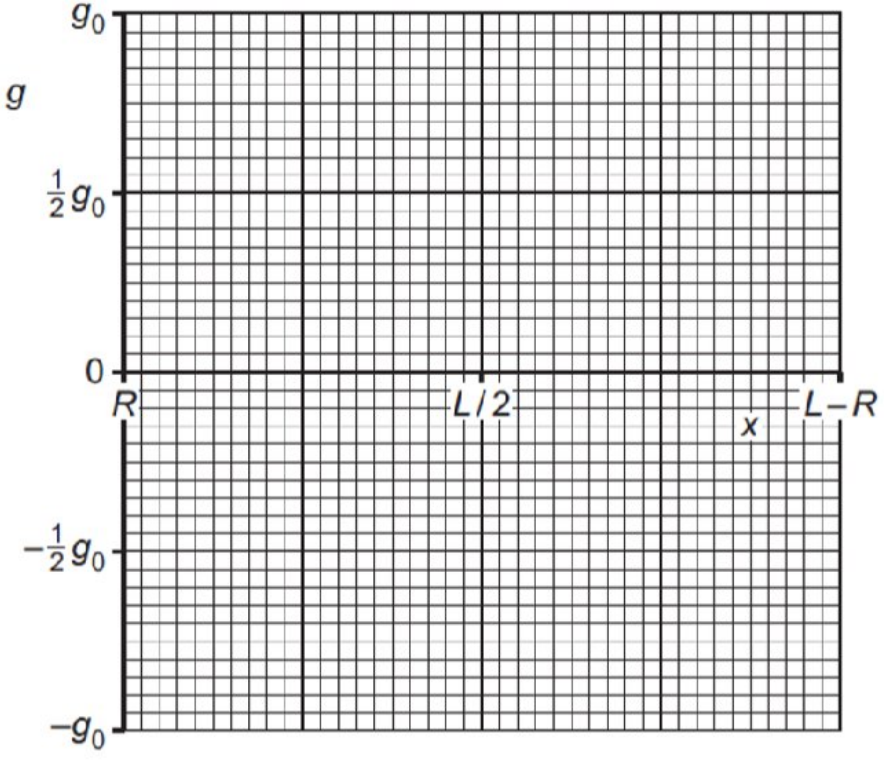


Fig. 3.3

[3]

[Total: 8]